



MTH101, Basic Linear Algebra

Monsoon 2025

16/10/2025, 2:00PM-3:00PM.

Tutorial-5

- (1) An $n \times n$ matrix A is called upper triangular if $a_{ij} = 0$ for $1 \leq j < i \leq n$. Using the principle of mathematical induction, show that if $A = (a_{ij})$ is an upper triangular matrix then $\det(A) = a_{11}a_{22} \cdots a_{nn}$.
- (2) Let A be an $n \times n$ matrix and E be an $n \times n$ elementary matrix. Prove that $\det(EA) = \det(E) \det(A)$.
- (3) Let I_n be the $n \times n$ identity matrix and A be a matrix such that $A^k = 0$ for some positive integer¹ $k \geq 1$. Prove that $I_n + A$ is an invertible matrix.
- (4) Let A be an $n \times n$ matrix. Using the principle of mathematical induction, verify that if A has a zero column then $\det(A) = 0$.

¹A matrix A such that $A^k = 0$ for some positive integer k is called a *nilpotent* matrix.